Machine Learning Foundations

Lumori · Version 2

Learning objectives

1. Describe how training examples support prediction.

2. Explain why test data must remain separate from training data.

Learning from examples

A machine learning model learns patterns from examples. Its training procedure adjusts model parameters to reduce a chosen loss. The resulting model can then produce predictions for new inputs [1].

A training set provides the examples used to update parameters. A validation set supports model selection. The test set is reserved for the final estimate of generalization.

A gradient update

Gradient descent: $\theta \leftarrow \theta - \eta \nabla L(\theta)$; a sum may be written as $\sum x_i$.

The learning rate η controls the size of each update. A lower training loss does not by itself establish better performance on unseen data [1, 2].

```
for sample in training_data:
    prediction = model(sample.features)
    loss = criterion(prediction, sample.target)
    parameters = parameters - learning_rate * gradient(loss)
```

Remember the distinction

Training optimizes the model. Evaluation estimates how well it performs on examples it has not used for training. Keep a record of the split and the metric used [2].

References

[1] Learning from data · p. 4

Course teaching team

https://example.edu/course/learning-from-data

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[2] 模型评估与数据划分 · p. 12

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